

Problem B. Matchmaker

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 128 megabytes

“Because of me, 60 couples are now happily married or engaged. I am a good matchmaker.”

– NorthStar English Book 2, 4th Edition

Mary is an expert of matchmaking. Every day, she needs to analyze the data and make possible matches. As she is more and more famous, the data have become too massive for Mary to calculate! Then, Mary comes for your help.

The mysterious method is described as follows:

Assume that there are N residents living in the city, numbered 1 to N , each with a value $A[i]$, which stands for the **lovesickness level** of resident i . For each resident x , the **willingness value** of x (**sender**) to request a date from another resident y ($y \neq x$) (**receiver**) is $(2022202227 \times A[x] + 2022202263 \times A[y]) \bmod 2022202299$.

Now, your job is to report the maximum willingness value of requesting a date between two people that are both numbered in the given range $[L, R]$, so that Mary can finish her calculations. That is, to find $\max\{(2022202227 \times A[i] + 2022202263 \times A[j]) \bmod 2022202299, \forall i, j \in [L, R]\}$.

Still, residents are very affected by their emotions. Hence, residents' lovesickness level $A[i]$ can vary along the time. You are also required to handle queries from Mary after some modifications. Are you up to the challenge?

Input

There are multiple test cases. The first line of input contains an integer T , indicating the number of test cases. For each test case:

The first line contains two integers N ($1 \leq N \leq 5 \times 10^5$) and M ($1 \leq M \leq 5 \times 10^5$), denoting the number of residents in the city and the number of queries.

The second line contains N integers $A[i]$ ($0 \leq A[i] \leq 10^4$), indicating the lovesickness level of resident i .

The following M lines contain three integers indicating M operations, and each operation is either **MODIFY** or **QUERY**:

- For **MODIFY**, the line contains 3 integers 0, x and v ($1 \leq x \leq N, 0 \leq v \leq 10^4$), which is to change the lovesickness level of resident x to v ($A[x] := v$).
- For **QUERY**, the line contains 3 integers 1, L and R ($1 \leq L \leq R \leq N$). Please report the the maximum willingness value of requesting a date between two people that are both numbered in the given range $[L, R]$.

It's guaranteed that the sum of N and the sum of M in all test cases will not exceed 5×10^5 .

Output

For each test case output one line containing one or three integer(s).

If there exist one or more requests, output the maximum willingness value of requesting a date between two people, the request sender, and the request receiver. If there exists multiple answer, output the answer with the **smallest** number denoting the sender, then the **smallest** number denoting the receiver.

If there is no valid pair of requesting a date, output **-1**.

Examples

standard input 1	standard output 1
2	2022202155 1 2
2 3	2022201759 2 1
1 2	-1
1 1 2	2022202191 1 2
0 1 11	
1 1 2	
2 2	
1 1	
1 1 1	
1 1 2	

standard input 2	standard output 2
1	2022201507 10 13
13 6	2022201543 10 6
1 11 19 18 14 9 10 12 18 6 18 11 10	2022201831 1 2
1 10 13	2022201579 10 11
1 4 12	
1 1 3	
0 2 3	
0 11 8	
1 7 13	

Note

In the first sample test set:

The first sample test case is explained below:

Operation	$A[1]$	$A[2]$	Output	Explanation
Initial state	1	2	/	
QUERY $[1, 2]$	1	2	2022202155 1 2	Obtained by resident 1 sending request to resident 2.
MODIFY $A[1] := 11$	11	2	/	Modify the lovesickness level of resident 1 to 11.
QUERY $[1, 2]$	11	2	2022201759 2 1	Obtained by resident 2 sending request to resident 1.

The second sample test case is explained below:

Operation	$A[1]$	$A[2]$	Output	Explanation
Initial state	1	1	/	
QUERY $[1, 1]$	1	1	-1	No possible request exists.
QUERY $[1, 2]$	1	1	2022202191 1 2	Either $(1 \rightarrow 2)$ or $(2 \rightarrow 1)$ can reach the same result. Since the pair $(1, 2)$ is lexicographically smaller than the pair $(2, 1)$, we should output $(1, 2)$.

This is a planet without gender. Love could be unequal and unilateral.

“Whether love is returned or not, every love that is true deserves to be seen, and it exists freely across all genders – including between two men, two women, or any hearts that choose each other.”

— Love is love, revised with pride, 2026. 